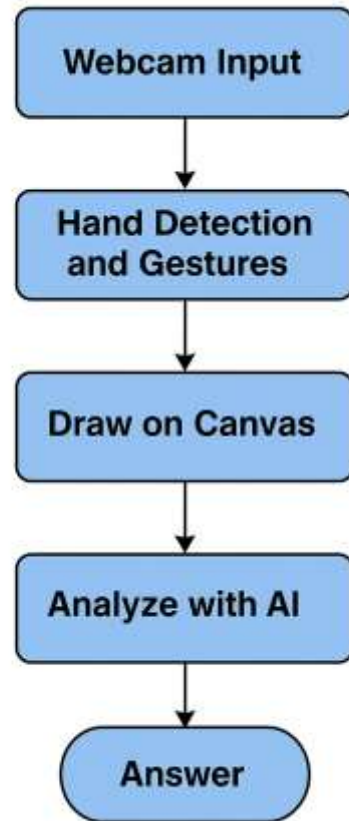




GESTURE BASED HANDWRITTEN MATH SOLVER USING AI & COMPUTER VISION



Project Overview

Build an interactive System :

- Detect hand gesture
- Draw math problems in air
- Solve them using Gemini AI

Combines

- Hand tracking(CVZone + OpenCV)
- Real time video processing
- Generative AI(GEMINI)
- Web UI(Streamlit)

Hand Gestures and Controls

Gesture



Function

Handwriting in air

Clearing the canvas

Submitting the math problem

Core Code Highlights

Hand Detection :

```
hands, img = detector.findHands(img)fingers =  
detector.fingersUp(hand)
```

Drawing Logic:

```
if fingers == [0,1,0,0,0]:  
  
    cv2.line(canvas, current_pos,  
prev_pos, (255, 0, 255), 10)
```

This code checks if only the index finger is up ([0,1,0,0,0]). If true, it draws a line on the canvas between the current and previous fingertip positions using a purple color (255, 0, 255) and a thickness of 10 pixels — essentially enabling "air writing".

AI Integration

Canvas submission to Gemini AI:

```
if fingers == [1,1,1,0,0]:
```

```
    pil_image = Image.fromarray(canvas)r
```

```
    response = model.generate_content(["Solve  
this math problem", pil_image])
```

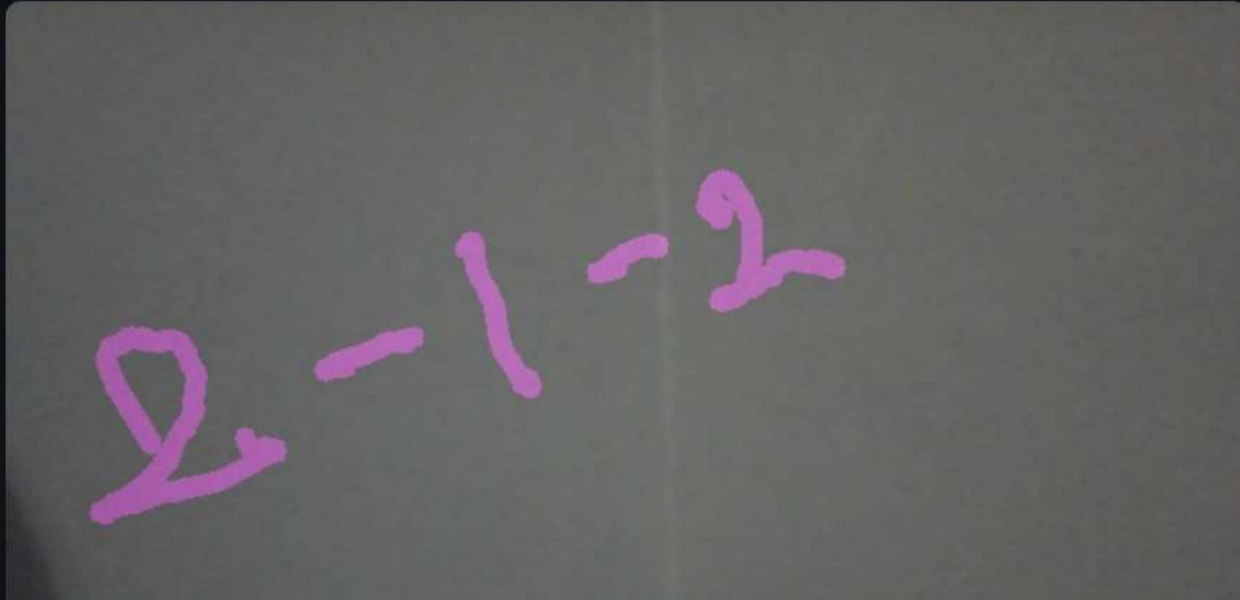
Receives solution and displays it in Streamlit UI

Select Mode

☒ Math Problem Solver

☐ System Control

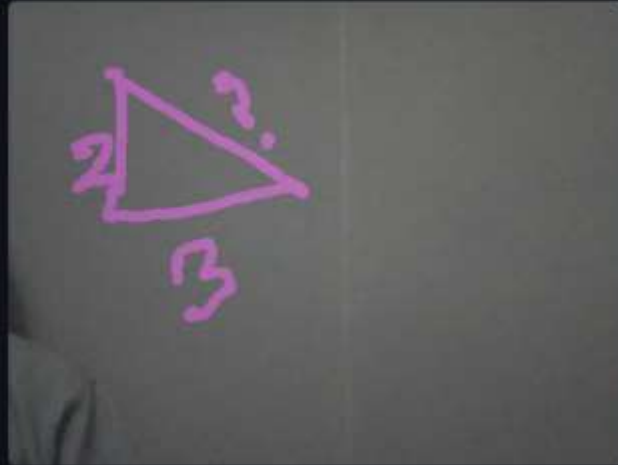
☒ Run



Output

The solution to $2 - 1 - 2$ is -1 .

Run



Output

The image shows a right-angled triangle. One leg is labeled as having length 2, the hypotenuse is labeled as having length 3, and the other leg is unknown (labeled with a question mark).

To find the length of the unknown leg, we can use the Pythagorean theorem: $a^2 + b^2 = c^2$, where a and b are the lengths of the legs and c is the length of the hypotenuse.

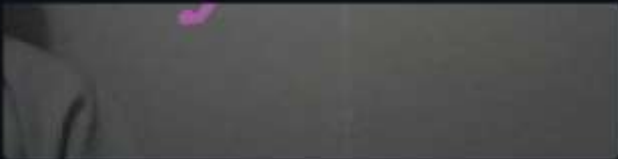
In this case:

* $a = 2$
* $c = 3$
* $b = ?$ (the unknown leg)

So we have:

$$2^2 + b^2 = 3^2$$

$$4 + b^2 = 9$$



BURNING... Stop Deploy

* $a = 2$
* $c = 3$
* $b = ?$ (the unknown leg)

So we have:

$$2^2 + b^2 = 3^2$$

$$4 + b^2 = 9$$

$$b^2 = 9 - 4$$

$$b^2 = 5$$

$$b = \sqrt{5}$$

Therefore, the length of the unknown leg is $\sqrt{5}$.

☒ Run

 RUNNING... **Stop** Deploy

Output

Here's how to solve the math problem $2^2 - 3^2$:

1. ****Calculate the squares:****

$$2^2 = 2 \cdot 2 = 4$$
$$\cdot 3^2 = 3 \cdot 3 = 9$$

2. ****Subtract:****

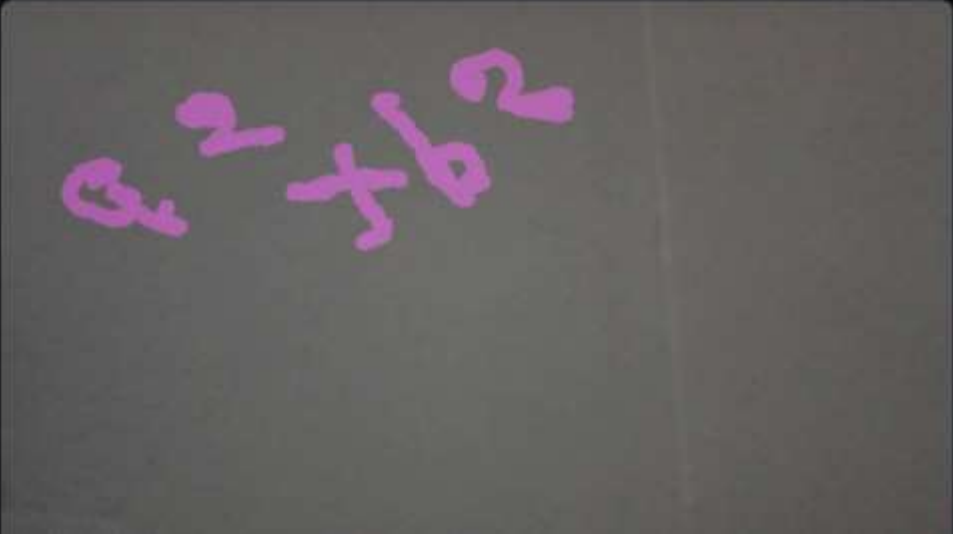
$$* 4 - 9 = -5$$

Therefore, the answer is $\boxed{-5}$

System Control

☒ Run

RUNNING... Stop Deploy



Output

That's an expression, not a problem to solve. $a^2 + b^2$ is the sum of the squares of a and b . There's no equation to solve, and therefore no single answer. To get a numerical answer, you would need values for a and b .

THANKYOU !